Recent Quiz Problems Covering Fall 26 Quiz 4 Material from Weeks 8 and 9

1. Compute the value of the integral:
$$\int_{\frac{\pi}{2}}^{\pi} 99\sin^3(x)\cos^8(x)\,dx$$

2. Compute the value of the integral
$$\int_0^{\pi} 9\sin^3(3x)\,dx$$

3. Compute the value of the integral
$$\int_0^{\pi} 40\sin^4(x)\,dx$$

4. Compute the value of the integral:
$$\int_0^{\frac{\pi}{4}} 162\tan^5(x)\sec^2(x)dx$$

5. Compute the value of the integral
$$\int_0^{\pi/3} 30\tan^5(x)\sec(x)\,dx$$

6. Compute the value of the integral
$$\int_0^{\pi/3} 15\sec^4(x)\tan^2(x)\,dx$$

7. Compute the value of the integral
$$\int_0^{\pi/6} 20\cos(3x)\cos(2x)\,dx$$

8. Compute the value of the integral using trig substitution:
$$\int_0^{\sqrt{3}} \frac{3x^3}{\sqrt{1+x^2}}\,dx$$

9. Compute the value of the integral using trigonometric substitution:
$$\int_0^{\sqrt{3}/2} \frac{24x^3}{\sqrt{1-x^2}}dx$$

10. Compute the value of the integral below using trig. substitution.
$$\int_{1/2}^{1} \frac{\sqrt{1-x^2}}{x^2}dx$$

11. Compute the value of the integral below using trig. substitution.

$$\int_{\sqrt{2}}^{\sqrt{5}} \frac{3x^3}{\sqrt{x^2-1}} dx$$

12. Compute the value of the integral

$$\int_0^{1/5} \frac{40}{(25x^2+1)^2}\, dx$$

13. Compute the Gini index of an economy given that its Lorentz curve is $f(x) = xe^{x-1}$.

14. Consider the region $\mathcal{R}$ in the first quadrant bounded by the following two functions:

$$f(x) = x^2 \ln(x)$$

$$g(x) = x^2$$

    (a) Find the points of intersection for the two functions to determine the bounds of integration. *(Note: You may assume the lower bound is the limit as $x \to 0^+$.)*

    (b) Set up a definite integral to find the area of the region $\mathcal{R}$.

    (c) Evaluate the integral.

15. Find the area of the region bounded by $y = \sqrt{4+x^2}$, the x-axis, $x = 0$, and $x = 4$.

16. Find the volume of the solid generated by rotating the region bounded by $y = \frac{1}{2}e^{x^2}$, $x = 0$, and $y = 0$ for $0 \le x \le \sqrt{\ln 5}$ about the y-axis.

17. Consider the region bounded on the right by the semicircle $x = \sqrt{4-y^2}$ and on the left by the line $x = 1$. Draw the region with a typical rectangle inside. Find the volume of the solid formed by revolving this region about the y-axis.

18. Consider the region bounded by $y = \sin(x)$ and $y = \cos(x)$ on the interval $[0, \pi/4]$. Draw the region with a typical rectangle inside. Find the volume of the solid formed by revolving this region about the x-axis.

19. Consider the region bounded by $y = 0, x = 0, y = e^x$ and $x = \ln(2)$.
Draw the region with a typical rectangle inside. Find the volume of the solid formed by revolving this region about the line $y = 2$.

20. Find the volume of the solid generated by the revolution of the region between the curves $y = 3\sqrt{x}$, $y = 3$ and $x = 0$ about the y-axis. The answer is of the form $\frac{a\pi}{b}$. Find a and b.

21. Find the volume of the solid generated by revolving the region bounded by $y = \frac{x^2}{\sqrt{9-x^2}}$, $y = 0$, and $x = 2$ about the y-axis.

22. **Setup but DO NOT solve.** Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x+3}$, $y = 0$, and $x + y = 3$ about the line $y = -2$.

23. Find the volume of the solid generated by revolving the region bounded by
$y = x$, $y = -x$, and $x = 2$, about the line $x = -3$.

24. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$, $x = 0$, and $y = 1$ about the line $y = -1$.

25. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}(x^2 + 16)^{1/4}$, $x = 3$, and $y = 0$, about the x-axis.